

YEAR 7 MATHEMATICS · CHAPTER 12

Ratios, rates and time

12.01 Ratios

12.03 Ratio problems

12.05 Best buys

12.07 Travel graphs

12.09 12-hour and 24-hour time

12.11 Timetables

12.02 Equivalent and simplified ratios

12.04 Rates

12.06 Rate problems

12.08 Time

12.10 Time differences

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

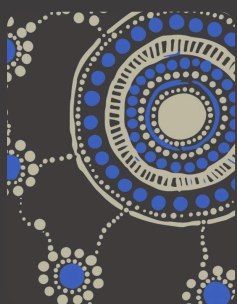
Contents

Lesson	Topic	Page	Done
12.01	Ratios	3	☐
12.02	Equivalent and simplified ratios	7	☐
12.03	Ratio problems	11	☐
12.04	Rates	15	☐
12.05	Best buys	19	☐
12.06	Rate problems	23	☐
12.07	Travel graphs	27	☐
12.08	Time	31	☐
12.09	12-hour and 24-hour time	35	☐
12.10	Time differences	39	☐
12.11	Timetables	43	☐

In this chapter you will

- write, simplify and use ratios
- solve ratio problems
- write rates and find the best buy
- solve rate problems and read travel graphs
- round, convert and calculate with time
- use 12-hour and 24-hour time, time differences and timetables

NSW Mathematics K-10 Syllabus (2022), Stage 4 Number and algebra: MA4-RAT-C-01 solves problems involving ratios and rates, and analyses distance-time graphs; time calculations from Stage 3 extended.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
12.01

Ratios

Date _____

Learning intentions

We are learning to:

- write a ratio to compare quantities

Success criteria

I can:

- ☐ write a ratio in the order the words give
- ☐ use a colon: 3 : 5
- ☐ find a part-to-part and a part-to-whole ratio

Key terms Write the definition as your teacher explains it.**Ratio****Part****Whole**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Ratio of shaded to unshaded?



Example 2

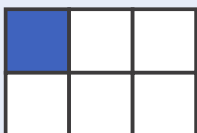
Ratio of shaded to unshaded?



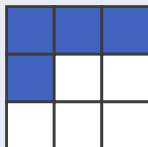
YOU DO

On your own: Set A. Write the ratio of shaded to unshaded.

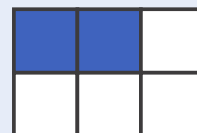
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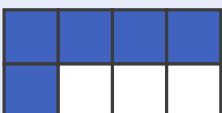
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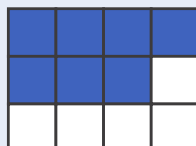
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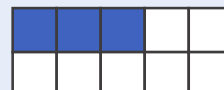
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4



6



YOU DO On your own: Set B. A class has 13 girls and 12 boys. Write the ratio of:

1 girls to boys

4 boys to the whole class

2 boys to girls

5 teachers (1) to students

3 girls to the whole class

6 students to teachers

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** 17 women and 38 men: find the ratio of men to women.**Example 4** Find the ratio of women to the total.**YOU DO** On your own: Set C. Write each ratio.

1 A bag has 7 red and 4 green apples: red to green

3 A team won 9, lost 5, drew 2: wins to losses

5 The letters in RATIO: vowels to consonants

2 Green apples to all apples

4 Losses to games played

6 A 9 m pole and a 4 m pole: short to long



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A recipe uses 2 cups of flour and 1 cup of sugar. Write the ratio of flour to sugar.
Example 6 Is $3 : 5$ the same as $5 : 3$? Explain.
YOU DO On your own: Set D. Think it through.**1** A drink is 1 part cordial to 4 parts water. What fraction is cordial?
3 Draw a 10-square diagram with a shaded to unshaded ratio of $3 : 2$.
2 In a ratio of cats to dogs of $3 : 2$, there are 12 cats. How many dogs?
4 Can a ratio be written as a fraction? Give an example.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

6 cats and 5 dogs. Write the ratio of cats to dogs.

Level 2

Write the ratio of dogs to all the animals.

Level 3

A juice is 1 part concentrate to 6 parts water. Write the ratio of water to the whole drink.

Extension · Stretch yourself

Worked example

The ratio of boys to girls is 4 : 5. What fraction of the class are boys?

1. Total parts = $4 + 5 = 9$
2. Boys are 4 of the 9 parts

Answer: $\frac{4}{9}$

YOU DO

On your own: read the worked example, then try these.

- E1** The ratio of red to blue to green beads is 2 : 3 : 5. What fraction are blue?

- E2** In a 3 : 7 ratio of juice to water, what percentage is juice?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A ratio compares quantities of the same kind.
 2. Write the parts in the order given.
 3. Use a colon: 3 : 5 (read “3 to 5”).
 4. Part to whole: add the parts for the whole.

Level 1

3 shaded, 5 unshaded

1. shaded : unshaded

Answer: 3 : 5

Level 2

Men to women

1. 38 : 17

Answer: 38 : 17

Level 3

Women to total

1. 17 : 55

Answer: 17 : 55

Extension

Boys : girls = 4 : 5

1. 4 of 9 parts

Answer: $\frac{4}{9}$ are boys

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LESSON
12.02

Equivalent and simplified ratios

Date _____

Learning intentions

We are learning to:

- find equivalent ratios and simplify ratios, including with units

Success criteria

I can:

- ☐ multiply or divide every part by the same number
- ☐ divide by the HCF to simplify
- ☐ change to the same units before simplifying

Key terms Write the definition as your teacher explains it.**Equivalent ratio****Simplest form**

Notes

Level 1 · Getting started**WE DO** With your teacher: copy what they write on the board.**Example 1** Complete: $2 : 3 = 8 : \underline{\quad}$ **Example 2** Complete: $2 : 3 : 4 = 6 : \underline{\quad} : \underline{\quad}$ **YOU DO** On your own: Set A. Complete each pair of equivalent ratios.

1 $2 : 3 = 8 : \underline{\quad}$

4 $5 : 2 = 15 : \underline{\quad}$

2 $1 : 5 = 2 : \underline{\quad}$

5 $2 : 3 : 4 = 6 : \underline{\quad} : \underline{\quad}$

3 $3 : 4 = \underline{\quad} : 20$

6 $10 : 15 = 2 : \underline{\quad}$

YOU DO On your own: Set B. Simplify.

1 12 : 21

4 25 : 45

2 25 : 75

5 18 : 6 : 24

3 6 : 36

6 5 : 25 : 100

Level 2 • Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Simplify 12 : 21.**Example 4** Simplify 18 : 6 : 24.**YOU DO** On your own: Set C. Change to the same units, then simplify.

1 2 cm : 10 mm

3 3 L : 600 mL

5 40 cents : \$2

2 5 dozen eggs : 6 eggs

4 50 m : 1 km

6 18 months : 3 years



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Simplify 2 cm : 10 mm.
Example 6 Simplify 20 minutes : 1 hour.
YOU DO On your own: Set D. Simplify.**1** 20 minutes : 1 hour
3 4 days : 10 hours
2 12 days : 3 weeks
4 1.5 : 2.5 (hint: $\times 2$ first)

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Complete: $3 : 7 = 9 : \underline{\quad}$ **Level 2**

Simplify 16 : 40.

Level 3

Simplify 30 cm : 2 m.

Extension · Stretch yourself

Worked example

Simplify $\frac{1}{2} : \frac{1}{3}$.

1. Multiply both by 6 (the LCD)
2. $3 : 2$

Answer: $3 : 2$

YOU DO

On your own: read the worked example, then try these.

E1 Simplify $\frac{3}{4} : \frac{5}{8}$.

E2 Simplify $0.25 \text{ kg} : 750 \text{ g} : 1.5 \text{ kg}$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Equivalent: $\times$ or $\div$ every part by the same number.
2. Simplify: $\div$ by the HCF of all parts.
3. Units: change to the same unit first, then drop the units.
4. Decimals or fractions: multiply to make whole numbers.

Level 1

$2 : 3 = 8 : ?$

1. $\times 4$

Answer: $8 : 12$

Level 2

$12 : 21$

1. $\div 3$

Answer: $4 : 7$

Level 3

$2 \text{ cm} : 10 \text{ mm}$

1. $20 \text{ mm} : 10 \text{ mm}$

Answer: $2 : 1$

Extension

$\frac{1}{2} : \frac{1}{3}$

1. $\times 6$

Answer: $3 : 2$

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LESSON
12.03

Ratio problems

Date _____

Learning intentions

We are learning to:

- solve problems using ratios

Success criteria

I can:

- ☐ use the unitary method: find one part first
- ☐ find an unknown quantity from a ratio
- ☐ divide a quantity in a given ratio

Key terms Write the definition as your teacher explains it.

Unitary method

Share

Notes

Level 1 • Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Girls : boys = 4 : 3. There are 75 boys.
How many girls?
Example 2 Tyrone : Erin invested in the ratio 5 : 7.
Erin invested \$63 000. How much did
Tyrone invest?
YOU DO On your own: Set A. Find the unknown.

1 $3 : 4 = 12 : ?$

2 $2 : 5 = ? : 30$

3 $7 : 2 = 35 : ?$

4 $1 : 6 = 8 : ?$

5 $5 : 3 = ? : 12$

6 $4 : 9 = 20 : ?$

YOU DO On your own: Set B. Share each amount in the ratio given.

1 \$60 in 1 : 2

4 72 lollies in 5 : 3

2 \$100 in 2 : 3

5 \$1000 in 1 : 4

3 \$45 in 4 : 5

6 48 kg in 1 : 2 : 3

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Share \$60 in the ratio 1 : 2.**Example 4** Share \$450 in the ratio 4 : 5.**YOU DO** On your own: Set C. Solve.

1 Girls : boys = 4 : 3. There are 75 boys. How many girls?

3 Paint is mixed blue : white = 2 : 5. How much white for 6 L of blue?

5 Map scale 1 : 50 000. 3 cm on the map is how many km?

2 Tyrone : Erin = 5 : 7. Erin invested \$63 000. Find Tyrone's share.

4 Share \$450 in the ratio 4 : 5.

6 Cement : sand : gravel = 1 : 2 : 3. How much sand in 36 kg?



Level 3 • Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A map has scale 1 : 50 000. 3 cm on the map is how far in real life?
Example 6 Cement, sand and gravel are mixed 1 : 2 : 3. How much sand is in 36 kg of mix?
YOU DO On your own: Set D. Harder problems.

- 1 The ratio of adults to children at a show is 3 : 8. There were 440 people. How many were children?

- 3 A recipe for 4 people uses 300 g of pasta. How much for 10 people?

- 2 Sam and Jo share money 3 : 5. Jo gets \$24 more than Sam. How much does each get?

- 4 A photo 10 cm by 15 cm is enlarged in the ratio 1 : 3. Find the new size.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Complete: $2 : 7 = 10 : \underline{\hspace{2cm}}$

Level 2

Share \$80 in the ratio 3 : 5.

Level 3

Juice : water = 1 : 4. How much water for 250 mL of juice?

Extension · Stretch yourself

Worked example

The ratio of red to blue marbles is 3 : 5. After 6 red marbles are added, the ratio is 1 : 1. How many blue marbles are there?

1. Red 3k, blue 5k
2. $3k + 6 = 5k$, so $2k = 6$, $k = 3$

Answer: 15 blue

YOU DO

On your own: read the worked example, then try these.

- E1** The ratio of boys to girls is 2 : 3. After 4 boys join, it is 1 : 1. How many girls?

- E2** Share \$84 among A, B and C so that A gets twice B and C gets three times B.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Find the value of one part.
 2. Multiply to find the other quantities.
 3. To share: total parts, then amount $\div$ total parts.
 4. Check the parts add back to the total.

Level 1

Girls : boys = 4 : 3, boys 75

1. 1 part = 25

Answer: 100 girls

Level 2

Tyrone : Erin = 5 : 7

1. 1 part = $63\,000 \div 7 = 9000$

Answer: \$45 000

Level 3

\$450 in 4 : 5

1. 9 parts, 1 part = \$50

Answer: \$200 and \$250

Extension

Marbles 3 : 5 $\rightarrow$ 1 : 1

1. $2k = 6$

Answer: 15 blue

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LESSON
12.04

Rates

Date _____

Learning intentions

We are learning to:

- write and simplify rates

Success criteria

I can:

- ☐ know a rate compares quantities of different kinds
- ☐ write a rate with both units, e.g. km/h
- ☐ simplify a rate to a rate per 1 unit

Key terms Write the definition as your teacher explains it.**Rate****Unit rate**

Notes

Level 1 · Getting started**WE DO** With your teacher: copy what they write on the board.**Example 1** Write as a rate: 150 000 voters for 50 politicians.**Example 2** Write as a rate: 1000 sheep over 5 hectares.**YOU DO** On your own: Set A. Write each as a rate in simplest form.**1** 150 000 voters for 50 politicians**4** 810 stitches for 3 balls**2** 1000 sheep over 5 hectares**5** \$120 for 8 hours**3** 10 tickets for \$8**6** 240 km in 3 hours

YOU DO On your own: Set B. Write each as a rate per 1 unit.

1 \$15 for 3 kg

2 60 L in 4 minutes

3 450 words in 5 min

4 72 beats in 1 min (per second)

5 \$4.50 for 5 L

6 56 km on 4 L

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Write as a rate: 10 tickets for \$8.**Example 4** Write as a rate: 810 stitches to make 3 balls.**YOU DO** On your own: Set C. Find each rate. Include units.

1 A car travels 240 km in 3 hours. Speed?

3 A worker earns \$187 for 8.5 hours. Hourly rate?

5 A printer prints 150 pages in 6 min. Pages per minute?

2 A tap fills 45 L in 9 minutes. Rate?

4 A plane flies 3600 km in 4.5 hours. Speed?

6 A runner runs 400 m in 80 s. Speed in m/s?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A car travels 240 km in 3 hours. Find the speed in km/h.
Example 6 Convert 90 km/h to metres per minute.
YOU DO On your own: Set D. Converting rates.**1** Convert 90 km/h to metres per minute.
3 Convert \$18 per hour to cents per minute.
2 Convert 5 m/s to km/h.
4 Which is faster: 20 m/s or 70 km/h?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write \$36 for 4 kg as a rate per kg.

Level 2

A car travels 150 km in 2 h. Find its speed.

Level 3

Convert 60 km/h to metres per minute.

Extension · Stretch yourself

Worked example

Convert 72 km/h to m/s.

1. $72 \text{ km} = 72\,000 \text{ m}$; $1 \text{ h} = 3600 \text{ s}$
2. $72\,000 \div 3600$

Answer: 20 m/s

YOU DO

On your own: read the worked example, then try these.

- E1** Convert 15 m/s to km/h.

- E2** A snail moves at 3 mm/s. How many metres per hour is that?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **rate** compares different kinds of quantities.
 2. Write both units: km/h, \$/kg, L/min.
 3. Unit rate: divide to get “per 1”.
 4. Convert units on the top and bottom separately.

Level 1

150 000 voters, 50 politicians

1. $\div 50$

Answer: 3000 voters/politician

Level 2

10 tickets for \$8

1. $8 \div 10$

Answer: \$0.80/ticket

Level 3

240 km in 3 h

1. $240 \div 3$

Answer: 80 km/h

Extension

72 km/h to m/s

1. $72\,000 \div 3600$

Answer: 20 m/s

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LESSON
12.05

Best buys

Date _____

Learning intentions

We are learning to:

- compare prices to find the best buy

Success criteria

I can:

- ☐ find the unit price (price per 1 unit or per 100 g)
- ☐ change to the same units before comparing
- ☐ decide the best buy and give a reason

Key terms Write the definition as your teacher explains it.**Unit price****Best buy**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Dog biscuits: 400 g of Woof for \$5.20.
Find the price per 100 g.**Example 2** 1 kg of Chow for \$11.10. Find the price
per 100 g. Which is the better buy?**YOU DO** On your own: Set A. Find the unit price (price per 1 item or 1 unit).

1 4 apples for \$2.40

4 5 L for \$8.50

2 6 eggs for \$3.90

5 12 pens for \$9

3 2 kg for \$7

6 3 m for \$4.20

YOU DO On your own: Set B. Find the price per 100 g.

1 400 g for \$5.20

4 500 g for \$4.25

2 1 kg for \$11.10

5 750 g for \$6

3 250 g for \$3

6 2 kg for \$19

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Milk: 1 L of Dairy Farm for \$2.85 or 1250 mL of Milky for \$3.90. Which is better value?**Example 4** Juice: 2 L for \$4.60 or 3 L for \$6.60. Which is better value?**YOU DO** On your own: Set C. Which is the better buy? Show working.

1 400 g of Woof for \$5.20 or 1 kg of Chow for \$11.10

3 2 L for \$4.60 or 3 L for \$6.60

5 6 rolls for \$4.80 or 10 rolls for \$7.50

2 1 L for \$2.85 or 1250 mL for \$3.90

4 500 g for \$3.75 or 800 g for \$5.60

6 375 mL for \$1.50 or 1.25 L for \$4.50



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Why might someone not choose the best buy?
Example 6 Pens: 3 for \$4.20 or 5 for \$6.50. Which is cheaper per pen?
YOU DO On your own: Set D. Problems.**1** Rice: 1 kg for \$2.40, 2 kg for \$4.50, 5 kg for \$10.50. Which is the best buy?
3 Is buying in bulk always the best choice? Explain.
2 A 600 g jar costs \$4.20. A 1 kg jar costs \$6.50. How much do you save per 100 g with the bigger jar?
4 Find a best buy at a supermarket (online). Show the unit prices.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the price per kg: 3 kg for \$12.60.

Level 2

Which is better: 1 L for \$3 or 2 L for \$5.40?

Level 3

Which is better: 250 g for \$2.50 or 1 kg for \$9?

Extension · Stretch yourself

Worked example

A shop has “3 for \$10” or \$3.60 each. How much do you save buying 6?

1. 6 at \$3.60 = \$21.60
2. 2 lots of 3 for \$10 = \$20

Answer: \$1.60

YOU DO

On your own: read the worked example, then try these.

- E1** Shampoo: 400 mL for \$6 or 1 L for \$13.50. Which is better? By how much per 100 mL?

- E2** A “buy 2 get 1 free” deal on \$4.50 items: find the price per item.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Find the price for the same amount (per 1 or per 100 g).
 2. Change units first (1 kg = 1000 g, 1 L = 1000 mL).
 3. The lower unit price is the better buy.
 4. Consider waste, storage and need.

Level 1

Woof

1. $5.20 \div 4$

Answer: \$1.30/100 g

Level 2

Chow

1. $11.10 \div 10$

Answer: \$1.11/100 g: better

Level 3

Milk

1. \$2.85/L vs \$3.12/L

Answer: Dairy Farm

Extension

3 for \$10 vs \$3.60 each (6)

1. \$20 vs \$21.60

Answer: save \$1.60

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.06

Rate problems

Date _____

Learning intentions

We are learning to:

- solve problems using rates

Success criteria

I can:

- ☐ multiply a rate by a quantity to find a total
- ☐ divide by a rate to find how many
- ☐ round sensibly

Key terms Write the definition as your teacher explains it.**Rate****Unitary method**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Marie earns \$22.45 per hour. How much for 38 hours?

Example 2

Petrol costs \$1.79/L. How many litres can you buy for \$65 (1 d.p.)?

YOU DO

On your own: Set A. Use the rate to find the total.

1 \$22.45/h for 38 h

4 15 L/min for 8 min

2 80 km/h for 2.5 h

5 60 words/min for 12 min

3 \$3.50/kg for 4 kg

6 \$1.79/L for 40 L

YOU DO On your own: Set B. Find how many or how long.

1 \$65 at \$1.79/L (1 d.p.)

2 150 kg at 20 kg/ha

3 210 km at 70 km/h

4 360 pages at 24 pages/min

5 \$90 at \$7.50/h

6 1000 L at 40 L/min

Level 2 • Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Fertiliser is used at 20 kg/ha. How many hectares does 150 kg cover?**Example 4** A car travels at 80 km/h for 2.5 h. How far does it go?**YOU DO** On your own: Set C. Solve.

1 A tap drips 12 mL per minute. How many litres per day?

3 A cyclist rides at 24 km/h. How far in 45 minutes?

5 A heart beats 72 times a minute. How many beats in a day?

2 A plumber charges \$85 call-out plus \$95/h. Find the cost for 3.5 h.

4 Grass seed is spread at 35 g/m². How much for a 12 m by 8 m lawn?

6 A car uses 7.5 L/100 km. How much fuel for 360 km?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** How long does it take to travel 210 km at 70 km/h?
Example 6 A tap drips 12 mL per minute. How many litres in a day?
YOU DO On your own: Set D. Harder problems.**1** Two taps fill a tank: one at 20 L/min, the other at 30 L/min. How long to fill 1500 L together?
3 Ann types 45 words/min and Ben 60 words/min. How much longer does Ann take to type 1800 words?
2 A car uses 8 L/100 km. Petrol is \$1.85/L. Find the fuel cost of a 450 km trip.
4 Which is faster: 100 m in 12.5 s or 1 km in 2 min 30 s?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the pay for 6 h at \$24.50/h.

Level 2

How long to travel 180 km at 60 km/h?

Level 3

How many litres of petrol for \$50 at \$1.60/L?

Extension · Stretch yourself

Worked example

A train leaves at 9 a.m. travelling at 90 km/h. A car leaves the same place at 10 a.m. at 120 km/h on the same road. When does the car catch up?

- At 10 a.m. the train is 90 km ahead
- The car gains 30 km every hour
- $90 \div 30 = 3$ hours after 10 a.m.

Answer: 1 p.m.

YOU DO

On your own: read the worked example, then try these.

- E1** How far from the start does the car catch up?

- E2** Two people walk towards each other from 12 km apart at 4 km/h and 2 km/h. When do they meet?

Lesson summary

Missed the lesson? Start here.

Remember 1. Total = rate $\times$ amount.
3. Time = distance $\div$ speed.

2. How many = total $\div$ rate.
4. Check units match (h and min, L and mL).

Level 1

$$\$22.45/\text{h} \times 38 \text{ h}$$

1. 22.45×38

Answer: \$853.10

Level 2

$$\$65 \div \$1.79/\text{L}$$

1. $36.31\dots$

Answer: 36.3 L

Level 3

$$150 \text{ kg} \div 20 \text{ kg/ha}$$

1. 7.5

Answer: 7.5 ha

Extension

Car catches train

1. $90 \div 30 = 3 \text{ h}$

Answer: 1 p.m.

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
12.07

Travel graphs

Date _____

Learning intentions

We are learning to:

- read and interpret travel (distance–time) graphs

Success criteria

I can:

- ☐ read distance and time from a travel graph
- ☐ identify stops (flat sections) and the return journey
- ☐ find speed = distance ÷ time for a section

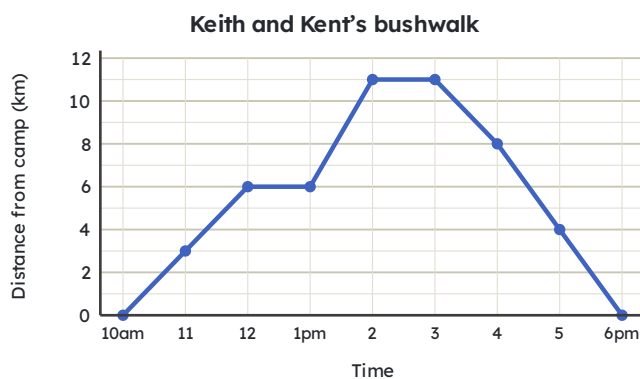
Key terms Write the definition as your teacher explains it.

Travel graph

Speed

Notes

Level 1 · Getting started

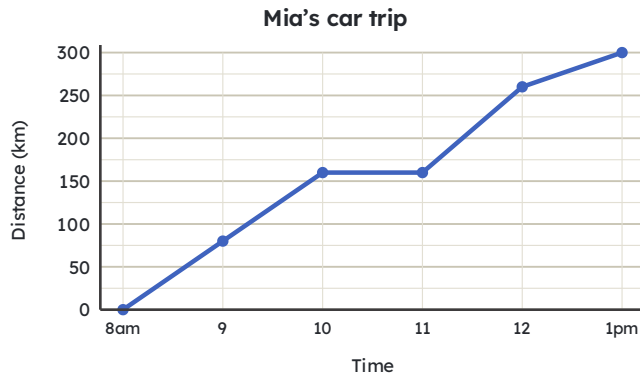
WE DO With your teacher: copy what they write on the board.**Example 1** Set A graph: how far did Keith and Kent walk altogether?**Example 2** How many stops did they make?**YOU DO** On your own: Set A. Use the travel graph.

1 Total distance walked?

2 Number of stops?

3 Time they started back?

4 Fastest section?

YOU DO On your own: Set B. Use the travel graph.

1 How far did Mia drive?

2 How long did she stop?

3 Speed from 8 to 10 a.m.?

4 Speed from 12 to 1 p.m.?

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** At what time did they start back?**Example 4** Between what times did they walk fastest?**YOU DO** On your own: Set C. Draw a travel graph for the journey.

1 Sam walks 4 km in 1 h, rests 30 min, walks 2 km in 30 min, then walks home (6 km) in 1.5 h. Start at 9 a.m.

Distance from home (km) against time

**Level 3 · Challenge**

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Calculate their speed in the last 2 hours.
Example 6 What does a steeper line mean on a travel graph?
YOU DO On your own: Set D. Think it through.**1** On a travel graph, what does a horizontal line show?
3 Write a story for Mia's car trip in Set B.
2 What does a line going down to the time axis show?
4 Mia's average speed for the whole trip (including the stop)?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What does a flat section on a travel graph mean?

Level 2

A graph goes from 0 km at 1 p.m. to 30 km at 3 p.m. Find the speed.

Level 3

In Set A, how far from camp were they at 11 a.m.?

Extension · Stretch yourself

Worked example

Two cyclists leave town A at 8 a.m. One rides at 20 km/h, the other at 25 km/h. How far apart are they at 10:30 a.m.?

1. Time = 2.5 h
2. Distances 50 km and 62.5 km

Answer: 12.5 km

YOU DO

On your own: read the worked example, then try these.

- E1** Draw both cyclists on one travel graph from 8 to 10:30 a.m.

- E2** When is the faster cyclist 10 km ahead?

Lesson summary

Missed the lesson? Start here.

Remember 1. Time is across, distance is up.
3. Steeper line means faster.

2. A flat line means stopped.
4. Speed = change in distance ÷ change in time.

Level 1

Total walked

1. 11 km out, 11 km back

Answer: 22 km

Level 2

Stops

1. Two flat sections

Answer: 2

Level 3

Last 2 hours

1. $8 \text{ km} \div 2 \text{ h}$

Answer: 4 km/h

Extension

Cyclists at 10:30

1. $62.5 - 50$

Answer: 12.5 km

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LESSON
12.08

Time

Date _____

Learning intentions

We are learning to:

- round, convert, add and subtract time

Success criteria

I can:

- ☐ round time to the nearest hour or minute
- ☐ convert between hours, minutes and seconds
- ☐ add and subtract hours and minutes, carrying 60

Key terms Write the definition as your teacher explains it.**Minute****Second**

Notes

Level 1 · Getting started**WE DO** With your teacher: copy what they write on the board.**Example 1** Round 9 h 50 min to the nearest hour.
Example 2 Round 3.2 h to the nearest hour. How many minutes is 0.2 h?
YOU DO On your own: Set A. Round to the nearest hour.**1** 9 h 50 min
4 7 h 30 min
2 3.2 h
5 1.75 h
3 4 h 12 min 49 s
6 12 h 29 min

YOU DO On your own: Set B. Convert.

1 316 min to h and min

4 3 h 15 min to min

2 208 s to min and s

5 0.4 h to min

3 2.5 h to min

6 1 day to min

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Convert 316 minutes to hours and minutes.**Example 4** Convert 208 seconds to minutes and seconds.**YOU DO** On your own: Set C. Evaluate.1 $6\text{ h }45\text{ min} + 3\text{ h }20\text{ min}$ 3 $4\text{ h }33\text{ min} - 2\text{ h }24\text{ min}$ 5 $1\text{ h }45\text{ min} \times 4$ 2 $3\text{ h }16\text{ min} + 1\text{ h }26\text{ min}$ 4 $2\text{ h }19\text{ min} - 1\text{ h }50\text{ min}$ 6 $7\text{ h }30\text{ min} \div 5$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** 6 h 45 min + 3 h 20 min**Example 6** 2 h 19 min – 1 h 50 min
YOU DO On your own: Set D. Problems.

- 1 A movie starts at 7:35 p.m. and runs for 2 h 18 min. When does it end?

- 3 A runner's lap times are 1 min 52 s and 2 min 07 s. Find the difference.

- 2 Three songs last 3 min 42 s, 4 min 15 s and 2 min 58 s. Find the total time.

- 4 Convert 2.85 h to hours and minutes.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 150 min to hours and minutes.

Level 2

Evaluate 2 h 40 min + 1 h 35 min.

Level 3

Evaluate 5 h 10 min – 2 h 45 min.

Extension · Stretch yourself

Worked example

How many seconds are there in a week?

1. $60 \times 60 \times 24 \times 7$

Answer: 604 800 s

YOU DO

On your own: read the worked example, then try these.

E1 About how many hours old are you?

E2 Convert 1 000 000 seconds to days, hours, minutes and seconds.

Lesson summary

Missed the lesson? Start here.

Remember 1. $60 \text{ s} = 1 \text{ min}$, $60 \text{ min} = 1 \text{ h}$, $24 \text{ h} = 1 \text{ day}$.
3. Add: add minutes, carry 60 min as 1 h.

2. Decimal hours: $0.1 \text{ h} = 6 \text{ min}$.
4. Subtract: borrow 1 h as 60 min.

Level 1

316 min

1. $316 \div 60 = 5 \text{ r } 16$

Answer: 5 h 16 min

Level 2

6 h 45 + 3 h 20

1. 9 h 65 min

Answer: 10 h 5 min

Level 3

2 h 19 – 1 h 50

1. 1 h 79 – 1 h 50

Answer: 29 min

Extension

Seconds in a week

1. $60 \times 60 \times 24 \times 7$

Answer: 604 800

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LESSON
12.09

12-hour and 24-hour time

Date _____

Learning intentions

We are learning to:

- convert between 12-hour and 24-hour time

Success criteria

I can:

- ☐ use a.m. and p.m. correctly
- ☐ write 24-hour time with 4 digits
- ☐ convert midnight and noon times correctly

Key terms Write the definition as your teacher explains it.

a.m.

p.m.

24-hour time

Notes

Level 1 · Getting started**WE DO**

With your teacher: copy what they write on the board.

Example 1

Convert 15:50 to 12-hour time.

Example 2

Convert 01:48 to 12-hour time.

YOU DO

On your own: Set A. Convert to 12-hour time.

1 01:48

2 06:24

3 15:50

4 12:30

5 23:05

6 00:40

YOU DO On your own: Set B. Convert to 24-hour time.

1 2:10 p.m.

4 1:50 p.m.

2 5:33 a.m.

5 11:45 p.m.

3 12:19 a.m.

6 12:05 p.m.

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Convert 12:30 to 12-hour time.**Example 4** Convert 2:10 p.m. to 24-hour time.**YOU DO** On your own: Set C. Use the bus timetable.

1 When does the first bus reach the Beach (12-hour time)?

2 How long does the trip from School to Beach take?

3 You must be at the Shops by 4 p.m. Which bus from School?

4 How often do the buses run?

Stop	Bus times (24-hour)		
School	15:10	15:40	16:10
Library	15:18	15:48	16:18
Shops	15:31	16:01	16:31
Beach	15:45	16:15	16:45

**Level 3 • Challenge**

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Convert 12:19 a.m. to 24-hour time.**Example 6** Why do airlines and hospitals use 24-hour time?**YOU DO** On your own: Set D. Problems.

- 1 A flight leaves at 22:40 and lands 3 h 35 min later. Write the landing time in 24-hour and 12-hour time.

- 3 Write the time 20 minutes before midnight in 24-hour time.

- 2 A shift starts at 18:30 and ends at 02:15. How long is it?

- 4 Which is later: 7:15 p.m. or 18:45?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Convert 17:25 to 12-hour time.

Level 2

Convert 8:05 p.m. to 24-hour time.

Level 3

Convert 12:45 a.m. to 24-hour time.

Extension · Stretch yourself

Worked example

A clock shows 09:40. What time will it be 1000 minutes later?

- 1000 min = 16 h 40 min
- $09:40 + 16:40 = 26:20 \rightarrow 02:20$ the next day

Answer: 02:20 (2:20 a.m.)

YOU DO

On your own: read the worked example, then try these.

- E1** What time is 500 minutes after 21:15?

- E2** How many minutes are there from 07:35 to 16:10?

Lesson summary

Missed the lesson? Start here.

Remember

- 24-hour time uses 4 digits: 00:00 to 23:59.
- p.m. times (except 12) add 12 hours.
- 12:xx a.m. is 00:xx; 12:xx p.m. is 12:xx.
- Subtract 12 from 13:00 onwards for p.m.

Level 1

15:50

1. $15 - 12 = 3$

Answer: 3:50 p.m.

Level 2

01:48

1. Before 12, a.m.

Answer: 1:48 a.m.

Level 3

2:10 p.m.

1. $2 + 12 = 14$

Answer: 14:10

Extension

1000 min after 09:40

1. + 16 h 40 min

Answer: 02:20

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LESSON
12.10

Time differences

Date _____

Learning intentions

We are learning to:

- calculate time differences and ages

Success criteria

I can:

- ☐ count on from the earlier time to the next hour, then to the later time
- ☐ work across midnight
- ☐ calculate an age in years, months and days

Key terms Write the definition as your teacher explains it.**Elapsed time****Time difference**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the time from 5:25 a.m. to 9:45 a.m.
Example 2 Find the time from 11:55 p.m. to 7:30 a.m.
YOU DO On your own: Set A. Find the time difference.

1 5:25 a.m. to 9:45 a.m.

4 10:40 a.m. to 1:05 p.m.

2 1:15 p.m. to 8:10 p.m.

5 06:15 to 11:50

3 07:50 to 14:20

6 9:35 a.m. to 12:00 noon

YOU DO On your own: Set B. Find the time difference (across midnight).

1 11:55 p.m. to 7:30 a.m.

4 10:20 p.m. to 6:05 a.m.

2 23:45 to 00:15

5 21:10 to 04:50

3 15:30 to 03:25 (next day)

6 8:30 p.m. to 2:15 a.m.

Level 2 • Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the time from 07:50 to 14:20.**Example 4** Find the time from 23:45 to 00:15.**YOU DO** On your own: Set C. Calculate.

1 Days from 25 March to 12 April

3 Weeks and days from 1 May to 30 June

5 Your age in years, months and days today

2 Days from 15 January to 1 March (not a leap year)

4 Samantha: born 3 June 2004. Age on 16 August 2023?

6 Time from 8:45 a.m. Monday to 3:20 p.m. Tuesday



Level 3 • Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Samantha was born on 3 June 2004.
Find her age on 16 August 2023.
Example 6 How many days from 25 March to 12 April?
YOU DO On your own: Set D. Problems.

- 1 A flight leaves Sydney at 9:50 p.m. and arrives in Perth at 12:45 a.m. local time. Perth is 2 h behind Sydney. How long is the flight?

- 3 Jack is 12 years 7 months old. His sister is 3 years 9 months younger. How old is she?

- 2 A baby was born on 28 February 2024. How old is she (in days) on 1 April 2024?

- 4 A parking meter charges \$3.20 per hour or part hour. Find the cost from 10:35 a.m. to 1:50 p.m.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the time from 8:20 a.m. to 11:05 a.m.

Level 2

Find the time from 22:30 to 01:10.

Level 3

How many days from 20 May to 4 June?

Extension · Stretch yourself

Worked example

It is 10:00 a.m. in Sydney. What time is it in London (9 hours behind)?

1. $10:00 - 9 \text{ h} = 1:00$

Answer: 1:00 a.m.

YOU DO

On your own: read the worked example, then try these.

E1 A video call is at 7:00 p.m. in Sydney. What time is it in London?

E2 Tokyo is 1 hour behind Sydney. A plane leaves Tokyo at 20:30 local time and takes 9 h 30 min. When does it land (Sydney time)?

Lesson summary

Missed the lesson? Start here.

Remember

1. Count on to the next hour.
2. Then count whole hours.
3. Then the extra minutes. Add them up.
4. For ages: years, then months, then days.

Level 1

5:25 to 9:45 a.m.

1. $35 \text{ min} + 3 \text{ h} + 45 \text{ min}$

Answer: 4 h 20 min

Level 2

11:55 p.m. to 7:30 a.m.

1. $5 \text{ min} + 7 \text{ h } 30 \text{ min}$

Answer: 7 h 35 min

Level 3

Samantha's age

1. $3 \text{ Jun } 2004 \rightarrow 3 \text{ Aug } 2023 \rightarrow 16 \text{ Aug}$

Answer: 19 y 2 m 13 d

Extension

10 a.m. Sydney

1. $- 9 \text{ h}$

Answer: 1:00 a.m.

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LESSON
12.11

Timetables

Date _____

Learning intentions

We are learning to:

- read and use timetables

Success criteria

I can:

- ☐ read departure and arrival times
- ☐ find travel times by subtracting
- ☐ choose the right service to arrive on time

Key terms Write the definition as your teacher explains it.**Timetable****Departure**

Notes

Level 1 • Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Train timetable (Set A): when does the 7:40 a.m. train reach Picton?**Example 2** How long does the 7:45 p.m. train take to get to Sydney?**YOU DO** On your own: Set A. Use the train timetable.

Station	a.m.	a.m.	a.m.	p.m.
Moss Vale	6:05	7:40	9:10	7:45
Bowral	6:12	7:47	9:17	7:52
Mittagong	6:20	7:55	9:25	8:00
Picton	6:58	8:33	10:03	8:38
Campbelltown	7:25	9:00	10:30	9:05
Sydney (Central)	8:21	9:56	11:26	10:01

- 1 When does the 6:05 a.m. train reach Sydney?

- 2 How long does it take?

- 3 When does the 9:10 a.m. train leave Mittagong?

- 4 How long from Picton to Sydney?

YOU DO On your own: Set B. Use the train timetable in Set A.

- 1 When does the 7:40 a.m. train reach Picton?
- 2 How long does the 7:45 p.m. train take to reach Sydney?
- 3 Latest train from Moss Vale to be in Sydney by 11 a.m.?
- 4 Bowral to Campbelltown on the 9:10 a.m. train?
- 5 Write the 7:45 p.m. arrival in 24-hour time.
- 6 How long between the first and second trains?

Level 2 • Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 What is the latest train you can catch at Moss Vale to be in Sydney by 11 a.m.?

Example 4 How long is the trip from Bowral to Campbelltown on the 9:10 a.m. train?

YOU DO On your own: Set C. Use the bus timetable.

Stop	Bus times (24-hour)		
School	15:10	15:40	16:10
Library	15:18	15:48	16:18
Shops	15:31	16:01	16:31
Beach	15:45	16:15	16:45

- 1 How long does it take from Library to Beach?
- 2 You arrive at the Library at 15:50. When is the next bus? When will you reach the Beach?
- 3 You need to be at the Shops by 4:30 p.m. What is the latest bus from School?
- 4 Write the times for the 16:10 bus in 12-hour time.

Finished Set D? Try the extension on the next page.

Example 6 Why are timetables often written in 24-hour time?

--

YOU DO On your own: Set D. Make your own timetable.

- 1 A ferry leaves the wharf every 40 minutes from 06:50 until 09:30. The trip takes 25 minutes. Write a timetable showing departure and arrival times.

Departs	Arrives

- 2 Your school day as a timetable: list the start and end of each period in 24-hour time.

Journal Pre-proof

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name	Class
------	-------



Level 1

When does the 9:10 a.m. train reach Campbelltown?



How long is the bus trip from School to Shops?

Level 3

Which train should you catch from Moss Vale to reach Sydney before 9 a.m.?

Extension · Stretch yourself

Worked example

You live 10 minutes' walk from Bowral station. You must be in Sydney by 10:15 a.m. What time must you leave home?

1. The 7:40 a.m. train arrives at 9:56 a.m.
2. It leaves Bowral at 7:47 a.m.
3. Leave home 10 min earlier

Answer: 7:37 a.m. (or earlier)

YOU DO

On your own: read the worked example, then try these.

- E1** Your friend at Picton wants to meet you on the same train. Which time does she board?

- E2** The 7:40 a.m. train is running 12 minutes late. When will it reach Sydney?

Lesson summary

Missed the lesson? Start here.

Remember

1. Find the row for the station and the column for the service.
2. Read across or down carefully.
3. Travel time = arrival – departure.
4. Choose the service that arrives before you need to be there.

Level 1

7:45 p.m. to Sydney

1. 7:45 p.m. → 10:01 p.m.

Answer: 2 h 16 min

Level 2

In Sydney by 11 a.m.

1. 9:10 arrives 11:26 (too late)

Answer: 7:40 a.m.

Level 3

Library to Beach

1. 15:18 → 15:45

Answer: 27 min

Extension

Leave home for Bowral

1. 7:47 – 10 min

Answer: 7:37 a.m.

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